

DRDO MALE UAV (TAPAS-BH-201 / RUSTOM-II) FLIGHT MANUAL & EMERGENCY SOP

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Aeronautical Development Establishment (ADE) / Indian Defence Standards (DEF-STAN)

1. MISSION THEATERS & REGIME PROFILES

1.1 Northern Sector (Ladakh High-Altitude Border Loiter)

- * **Operating Altitude:** 18,000 - 24,000 ft MSL (Flight Level FL180 - FL240).
- * **Base Station:** Leh Air Force Station (3,256 m MSL) / Nyoma Advanced Landing Ground.
- * **Environmental Stress:** Ambient temperature -20 deg C to -32 deg C. High density altitude reduces air cooling mass flow by up to 48%, significantly increasing cylinder thermal runaway risk during extended loiter.
- * **Standard Power Profile:** 72% TPS, 5,120 RPM, 90 KIAS, TAS 122 knots.

1.2 Western Sector (Thar Desert Extreme Heat Patrol)

- * **Operating Altitude:** 3,000 - 8,000 ft MSL.
 - * **Environmental Stress:** Ambient temperature +42 deg C to +49 deg C. Airborne sand particulates accelerate injector nozzle wear and air filter dust loading. High ambient temperatures reduce oil cooler thermal dissipation margin.
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2. DRDO CANONICAL FAILURE MODES & FAST SOP DIRECTIVES

2.1 Summary Matrix (PS-26054)

Fault ID	Alert Title	Primary Indicator	Severity	SOP Emergency Action
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01	Cylinder #2 CHT Overheat	CHT_2 > 135 deg C, Delta CHT > 25 deg C	CRITICAL	Enrich trim +12%, 4,600 RPM, descend 3,000 ft
02	Fuel Injector #1 Clog	Fuel Flow down 20%, EGT_1 > 880 deg C	CRITICAL	Switch FADEC Lane B, engage aux boost pump
03	Ignition Misfire (Lane A)	EGT_2 down 120 deg C, RPM Flutter	WARNING	Select Ignition Lane B, hold cruise power
04	Oil Pressure Decay	P_oil < 2.0 bar, T_oil elevated	CRITICAL	Reduce to 4,200 RPM, immediate RTB vector
05	Gearbox Vibration / Clutch	VIB > 3.2 mm/s, 3rd Harmonic	WARNING	Avoid 4,000-4,400 RPM resonance band
06	Exhaust EGT Imbalance	EGT_3 > 870 deg C, Delta EGT > 65 deg C	WARNING	Apply +6% individual fuel trim on Cyl #3
07	Alternator Voltage Sag	V_bus < 12.8 V, I_bat < -10 A	WARNING	Shed ISR radar/payload, monitor battery SOC
08	Dual FADEC ECU Drift	Delta MAP > 8 kPa, Fuel Flow up	WARNING	Force Lane B arbitration, log post-flight

3. MISSION READINESS (GO / NO-GO) CRITERIA

1. ****GO Status:**** All 27 telemetry parameters within nominal green bands. Autoencoder anomaly score < 0.35. RUL estimate > 12.0 hours.
2. ****CAUTION Status:**** Sub-threshold prognostic drift detected or non-critical advisory (e.g. minor EGT divergence or alternator voltage sag with adequate battery reserve). Mission may continue with enhanced telemetry monitoring.
3. ****NO-GO Status:**** Any CRITICAL fault (CHT overheat > 140 deg C, Oil pressure < 1.8 bar, Injector starvation with severe vibration). Autonomous return-to-base (RTB) or precautionary landing mandatory.